

Detection of Potential Depressive Disorders on “X” Social Media Based on User-Level Analysis Using Multi-Modal Transformers Architecture

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Abstract—Existing automated depression detection systems on social media are often limited to post-level analysis, failing to capture chronic emotional patterns. This study develops a user-level depression identification system on the “X” platform based on DSM-5 clinical criteria. A Multimodal Transformer architecture was developed by fine-tuning IndoBERTweet for text and Vision Transformer (ViT) for image feature extraction. The model analyzes users’ timeline histories to distinguish situational negative emotions from consistent clinical indications. Transparency is ensured through Explainable AI (XAI) attention maps. Implemented as a browser extension, the system aggregates 50-100 recent posts for hybrid scoring. Experimental results show the Multimodal ViT fusion achieves 82.10% accuracy and 79.10% F1-Score. Although the performance increase over the text-only model is not statistically significant, the multimodal fusion reduces false-negatives to the lowest observed count (13 vs. 16 for ResNet-18), a clinically critical advantage. Clinical psychologist validation confirmed alignment with medical reasoning and ethical output terminology (“Stress Indication”). Furthermore, End-User Testing yielded a System Usability Scale (SUS) score of 80, proving its feasibility as a proactive, user-friendly early screening instrument. These findings demonstrate that integrating multimodal deep learning with explainable AI presents a promising, ethically responsible decision-support tool for preliminary mental health screening. By prioritizing high recall, the system serves as a practical screening aid to facilitate proactive mental health outreach in digitized environments.

Keywords—Depression, IndoBERTweet, Social Media X, Explainable AI, Vision Transformer, DSM-5, Browser Extension.

I. INTRODUCTION

The rapid growth in the use of social media such as X (formerly Twitter) has fundamentally changed the global communication landscape, allowing users to express their thoughts, feelings, and emotional states instantly. The X platform remains one of the primary spaces frequently used by individuals to pour their hearts out (*self-disclosure*) anonymously, indirectly recording the digital footprint of their mental health conditions [1].

Mental health, particularly depressive disorders, has become a crucial issue requiring immediate attention globally and specifically in Indonesia. According to the World Health Organization (WHO), depression is a leading cause of disability worldwide, affecting over 280 million people [2]. In the con-

text of Indonesia, the Ministry of Health reports a significant surge in depressive symptoms among adolescents and young adults (Gen Z), heavily correlated with the post-pandemic digital lifestyle. Based on the clinical criteria in the *Diagnostic and Statistical Manual of Mental Disorders* (DSM-5), major depression is characterized by persistent symptoms such as depressed affect, constant loss of energy, and anhedonia, which adversely impact social functioning [3]. The DSM-5 explicitly notes that a diagnosis requires these symptoms to be present for a minimum of two weeks, emphasizing the critical need for temporal, long-term observation rather than isolated assessments. Although the urgency is real and its prevalence continues to rise, many depression sufferers are reluctant to seek professional help, primarily because of social stigma and limited access to psychological facilities [4].

These conditions drive sufferers to prefer expressing their despair through social media posts using unique language patterns. Psycholinguistic research shows that depression sufferers tend to use absolutist words [5] and an increased intensity of first-person singular pronouns [6]. Preemptive detection through social media analysis can provide opportunities for earlier clinical intervention [7].

However, the main challenge is the highly informal nature of social media language. In Indonesia, X users dominate the timeline with slang, non-standard abbreviations, code-mixing, and metaphorical expressions [8]. Furthermore, accurate depression detection must be conducted by collectively analyzing the user’s entire posting history (*user-level analysis*) to distinguish situational sadness from chronic depressive episodes [9], [10]. Existing multimodal studies largely ignore user-level temporal patterns, limiting their applicability to real-world early detection scenarios.

Another limitation is that conventional models often rely solely on text. In reality, users frequently post images that reinforce emotional intensity [11]. Unimodal analysis is prone to diagnostic errors when sentence meaning heavily depends on accompanying visual context.

Based on this background, this study designs an intelligent system for detecting potential depression at the user level using a Multimodal Transformers architecture. IndoBERTweet

is combined with the Vision Transformer (ViT) using late fusion to process complex text-image relations. An Explainable AI (XAI) mechanism is also integrated to visualize the reasoning behind each decision, ensuring clinically transparent and accountable results.

II. LITERATURE REVIEW

The X platform encourages open self-disclosure. The linguistic expression of depression sufferers is often characterized by high usage of first-person singular pronouns [6] and absolutist words (e.g., always, never) indicating dichotomous thinking [5]. In the Indonesian context, these expressions predominantly use non-standard language, abbreviations, and code-mixing [8].

Previous studies attempted to solve this challenge through various approaches. Al Asad et al. [9] proposed user-level analysis but did not integrate visual modality. Zogan et al. [10] combined textual and behavioral features to improve accuracy, yet their model remains text-only. Willie et al. [12] highlighted the urgency of domain-specific language models, though they did not address the Indonesian slang context. These text-only approaches exhibit a critical vulnerability when analyzing modern social media behavior, where depressive affect is frequently masked by sarcastic text or juxtaposed against contradictory imagery. A unimodal text system processing a user's self-deprecating meme would invariably misclassify it as non-depressive due to the absence of explicit distress vocabulary. Therefore, integrating a visual modality is no longer an optional enhancement but a theoretical necessity to capture the holistic emotional state of the user.

The Transformers architecture [13], which fundamentally revolutionized natural language processing by dispensing with recurrence and convolutions, relies entirely on a sophisticated attention mechanism. This architecture evolved into BERT [14], which gave rise to IndoBERTtweet, a domain-specific model trained on 409 million Indonesian tweets. Its Scaled Dot-Product Attention allows simultaneous computation of relational weights across words, capturing long-range dependencies and bidirectional context. This capability makes it the new gold standard for understanding the highly nuanced and informal Indonesian text on the X platform [8].

To address the visual modality, the Vision Transformer (ViT) adapts the same self-attention mechanism for computer vision by dividing images into a sequence of non-overlapping fixed-size patches ($P \times P$) [15]. By treating these patches similarly to tokens in NLP, ViT applies Positional Embedding to preserve the spatial understanding of objects. This purely self-attentional approach to images has proven highly robust, particularly when aligned with Transformer-based text models in cross-modal learning scenarios [16].

In multimodal systems, the fusion strategy dictates how different modalities interact. The late fusion strategy, fundamentally based on feature concatenation at the decision level [17], allows the text (IndoBERTtweet) and image (ViT) encoders to operate independently and optimize their distinct modality-specific weights. Their respective 768-dimensional outputs

are concatenated into a highly expressive 1536-dimensional multimodal vector. This approach avoids the feature distortion and overwhelming computational complexity that are highly prone to occur in early or mid-level fusion architectures [18].

An Explainable AI (XAI) module is implemented to visualize Attention Maps and dominant keyword contributions [19]. SHAP (SHapley Additive exPlanations) is crucial for opening the “black-box” of deep learning [20], providing transparent empirical evidence that clinical psychologists can verify.

III. METHODOLOGY

The overall system workflow is illustrated in Fig. 1.

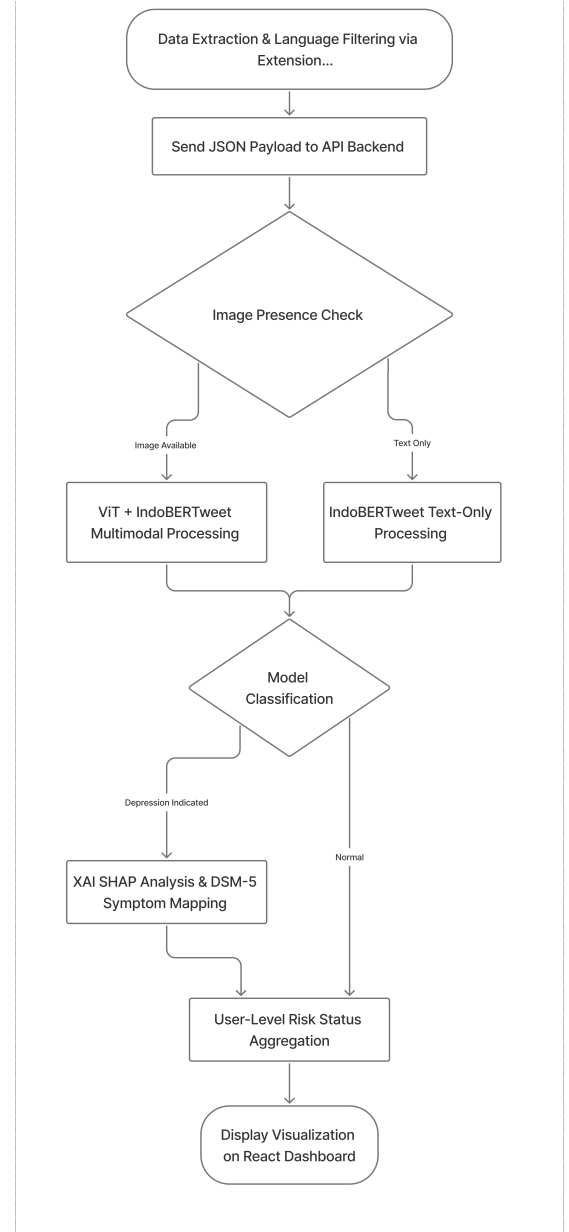


Fig. 1. System flowchart.

A. Data Collection and Inter-Rater Reliability (Gwet's AC1)

Data collection employed a dual-source strategy. The textual corpus originates from a public Indonesian Twitter dataset (6,982 raw tweets), augmented via the *llama-3.3-70b-versatile* LLM using tailored prompts to generate Gen Z slang variations and contextual depressive metaphors. To resolve class imbalance, dynamic undersampling was applied to the majority Normal class to match the expanded Depressive instances, yielding a strictly balanced benchmark corpus of 7,991 text samples (50% depressive, 50% normal). The corpus was partitioned with a 70/15/15 ratio into 70% training (5,593 samples), 15% validation (1,199 samples), and an independent 15% test set (1,199 samples). To prevent lexical data leakage, group-based splitting ensured that original tweets and their LLM paraphrased variations were strictly bound to the same partition and never crossed into the test split.

In parallel, a multimodal dataset comprising 294 organic tweet-image pairs was collected in real-time via Selenium WebDriver, filtered for Indonesian personal self-disclosures. To eliminate user-level identity and writing-style leakage, data points were sampled across distinct, non-overlapping user timelines (user-disjoint sampling). Three psychology student annotators independently labeled the multimodal corpus. Inter-Rater Reliability was measured using Gwet's AC1 [21] to avoid high-agreement paradoxes common to Cohen's Kappa, formulated as:

$$AC1 = \frac{p_a - p_e(\gamma)}{1 - p_e(\gamma)} \quad (1)$$

where p_a represents overall percent agreement and $p_e(\gamma)$ is the chance-agreement probability adjusted for marginal distributions. Across 195 overlapping cross-annotated instances, the evaluation yielded an AC1 score of 0.8558, demonstrating "Almost Perfect Agreement" under Gwet's benchmark guidelines. Ambiguous labels were resolved via consensus with a licensed clinical psychologist. To respect subject privacy while maintaining raw pixel fidelity for ViT feature extraction, this organic multimodal repository is strictly held in a private research vault.

B. Data Preprocessing

Text preprocessing for IndoBERTweet includes Data Cleansing (removing URLs, hashtags, mentions), Case Folding to lowercase, and Tokenization using the built-in IndoBERTweet tokenizer. Image preprocessing for ViT includes Resizing to 224×224 pixels, Normalization of pixel intensity, and Image Patching into 16×16 pixel patches.

C. Multimodal Transformer Architecture

The proposed architecture integrates the pre-trained IndoBERTweet [8] to extract text features (768-dimensional vector) and ViT to extract image features. The combination is carried out through Late Fusion (feature concatenation) [17], producing a 1536-dimensional combined vector v_{fusion} :

$$v_{fusion} = [v_{text} \oplus v_{image}], \quad v_{fusion} \in \mathbb{R}^{1536} \quad (2)$$

where $\oplus$ denotes the concatenation operator. A Dynamic Routing mechanism automatically switches to text-only fallback mode if no image is detected, ensuring lightweight inference.

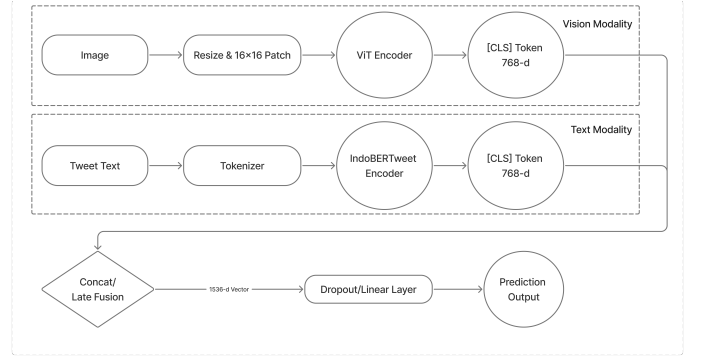


Fig. 2. Multimodal Late Fusion Architecture.

D. Recall-Priority Training Strategy

To minimize False Negatives, the training strategy prioritized Recall as the primary model selection metric via `metric_for_best_model='recall'` in the HuggingFace Trainer. Additionally, the sentence-level classification threshold was lowered from 0.50 to 0.25, so that any tweet with a depression confidence score above 25% is flagged as an indication. The fine-tuning ran for 3 epochs (IndoBERTweet) with a batch size of 16, learning rate 2×10^{-5} , and weight decay 0.01. All computational experiments and model training were conducted in a Python 3.10 environment utilizing the PyTorch framework and HuggingFace Transformers library. Hardware acceleration was provided by an NVIDIA T4 Tensor Core GPU with 16GB of VRAM, ensuring optimal matrix multiplication performance for the high-dimensional attention layers.

E. User-Level Hybrid Scoring Mechanism

While the deep learning models operate as the core post-level affective feature extractors, psychiatric diagnosis under DSM-5 guidelines mandates evaluating persistent emotional patterns across time rather than isolated posts. To bridge post-level inference into user-level screening, the system aggregates the chronological timeline history of 50–100 recent posts within a 14-day sliding window. Risk severity is calculated using the Hybrid Scoring method [22], [23], determining the final user-level risk score (R) by combining the global average confidence (S_{global}) and the peak average score (S_{peak}):

$$R = (0.4 \times S_{global}) + (0.6 \times S_{peak}) \quad (3)$$

This formulation emphasizes acute depressive episodes while maintaining the context of longitudinal mood consistency. The aggregated user risk (R) is categorized into High Potential (>30%), Moderate (15%–30%), and Stable (<15%).

F. Dashboard Implementation and Semantic Mapping (SBERT)

The final implementation is packaged within a Client-Server architecture: a browser extension for passive real-time scanning and DOM extraction, and a FastAPI backend. The system implements Indo-SBERT (Sentence-BERT) to extract semantic vectors from tweets and calculate Cosine Similarity against the nine main DSM-5 criteria. The semantic alignment score for each post d against DSM-5 criterion c_i is computed as:

$$\text{sim}(d, c_i) = \frac{\vec{v}_d \cdot \vec{v}_{c_i}}{\|\vec{v}_d\| \|\vec{v}_{c_i}\|} \quad (4)$$

where $\vec{v}_d$ and $\vec{v}_{c_i}$ are the SBERT embedding vectors for the tweet and the DSM-5 criterion respectively. A post is flagged as aligned with a criterion if $\text{sim}(d, c_i) > 0.35$. The system counts the number of distinct DSM-5 criteria triggered across the monitoring window, providing a structured clinical evidence trail for each flagged user. A React-based dashboard presents chronological emotional trend visualizations over a 14-day window, text highlighting, and early intervention recommendations [3].

IV. RESULTS AND DISCUSSION

A. Text Classification Model Testing Results

The pre-trained IndoBERTweet model was compared with standard IndoBERT and BiLSTM on an independent test set (15% of the data corpus). BiLSTM experienced extreme difficulty mapping implicit semantics, yielding an Accuracy of 83.46% and a concerningly low Recall of 69.20%, despite a paradoxically high Precision of 97.95%. This discrepancy is characteristic of a precision-recall tradeoff under class imbalance: the BiLSTM learned to predict the majority class (Normal) with high confidence, resulting in high Precision but catastrophically missing a large portion of true depressive cases (low Recall). The sequential nature of the BiLSTM architecture fundamentally struggled to capture the long-range dependencies and inverted syntax commonly found in Indonesian social media slang. When users expressed depressive thoughts using fragmented sentences or abruptly shifted context midway through a tweet, the BiLSTM’s memory gates often discarded critical early context. Conversely, the self-attention mechanism within Transformer-based architectures successfully overcame this limitation by computing global word relations simultaneously. Although standard IndoBERT recorded slightly higher baseline metrics, qualitative analysis revealed this over-performance was artificially inflated by test samples that were too formal. IndoBERTweet proved its true robustness and superiority against complex tweets using non-standard grammar (*jujurly, capek bgt, pen nyerah*), as illustrated in Table II. Detailed quantitative performance is provided in Table I.

B. Multimodal Model Testing Results

The multimodal architecture was evaluated using 5-Fold Cross Validation. The Multimodal ViT fusion achieved the

TABLE I
TEXT CLASSIFICATION MODEL PERFORMANCE COMPARISON

Model	Acc	Prec	Rec	F1
BiLSTM	83.46%	97.95%	69.20%	81.10%
IndoBERT (Standard)	98.33%	97.51%	99.28%	98.38%
IndoBERTweet	97.77%	97.48%	98.19%	97.83%

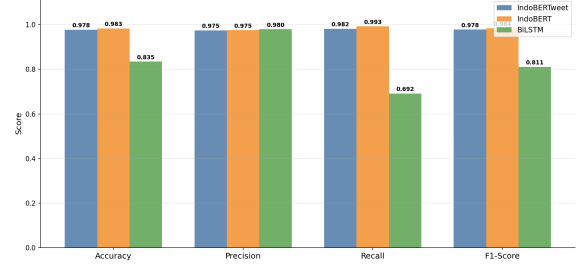


Fig. 3. Performance metrics comparison of text classification models.

TABLE II
SLANG LANGUAGE DETECTION COMPARISON

Tweet Sample	IndoBERT	IndoBERTweet
“jujurly gw capek bgt pen nyerah aja idup rasanya hftt”	Normal	Indicated
“males bgt anjir nugas mulu kapan re-bahannya wkwk”	Normal	Normal

highest Accuracy (82.10%) among all multimodal variants, outperforming Multimodal ResNet-18 (81.77%), as summarized in Table III. For reference, the text-only baseline achieved 81.10% Accuracy. Crucially, ViT suppressed False Negatives to only 13 cases (vs. 16 for ResNet), the absolute priority in a psychological diagnostic system.

TABLE III
MULTIMODAL CLASSIFICATION COMPARISON (5-FOLD CV) (BEST FALSE-NEGATIVE COUNT HIGHLIGHTED)

Architecture	Accuracy	False Neg.
Multimodal ResNet	81.77%	16
Multimodal ViT	82.10%	13

A Paired t-Test at $\alpha = 0.05$ confirmed that the performance increase is not statistically significant ($p > 0.05$ for all metrics, smallest $p = 0.345$ for Accuracy). Detailed significance test results are in Table IV. This null result is expected and attributable to three compounding factors. First, the multimodal dataset size of only 294 samples yields insufficient statistical power to detect small performance differences (approximately 1%) with a paired t-Test, a well-known limitation of small-sample clinical studies. Second, a *ceiling effect* is observed: IndoBERTweet already captures the majority of discriminative depressive linguistic signals, leaving a narrow margin for the visual modality to contribute additional predictive value. Third, and most critically, images on the X platform are inherently noisy; users frequently attach incongruent visual media (e.g.,

memes, song lyrics, or random aesthetic photos) to depressive texts, causing the ViT encoder to receive misleading rather than complementary affective signals. This is a platform-specific limitation, not an architectural failure. Despite the absence of statistical significance, the multimodal fusion was retained because it is clinically justified: minimizing False Negatives is the paramount objective in any mental health screening system, and the ViT fusion achieved the lowest False Negative count among all evaluated multimodal architectures.

TABLE IV
SIGNIFICANCE TEST RESULTS (PAIRED T-TEST)

Metric	Text	Multi	Diff	p-val	Sig.
Acc	0.811	0.821	+0.010	0.345	No
Prec	0.769	0.779	+0.010	0.482	No
Rec	0.818	0.825	+0.007	0.721	No
F1	0.781	0.791	+0.010	0.459	No

C. Clinical Psychologist Expert Consultation Results

The system’s clinical feasibility was evaluated through structured Expert Consultation with a licensed Clinical Psychologist via system demonstration and an in-depth qualitative interview. The evaluation covered three main aspects: (1) *System Functionality Validation*, assessing the relevance of using social media digital footprints as a psychological window and whether the analytical dashboard output is beneficial for accelerating initial screening; (2) *Accuracy and Medical Logic Validation*, evaluating the alignment of computational parameters with clinical psychology standards, including risk threshold validation, XAI keyword accuracy, DSM-5 symptom mapping, and sarcasm handling in multimodal data; and (3) *Temporal Observation and Efficiency Validation*, assessing the temporal visualization feature’s ability to differentiate reactive emotional fluctuations from persistent depression under DSM-5 criteria, and evaluating the ethical boundaries of automated data collection. The expert confirmed positive results across all three dimensions. The data fetching flow was considered reliable, and the language criteria mapping aligns with standard clinical indicators.

The expert particularly appreciated the SHAP Partition Explainer module for text (Fig. 4). As validated by the expert: “*The visual explanations increase trust and the stress-labeling avoids harmful self-diagnosis.*” As an ethical mitigation, the system output uses the label “Indicated” (instead of “Depression”) to prevent the nocebo effect of self-diagnosis. This expert feedback confirms the qualitative feasibility of the system as an ethical and responsible preliminary screening and decision-support aid, rather than an autonomous diagnostic instrument.

D. End-User Acceptance Testing (UAT) Results

End-user testing adopted the System Usability Scale (SUS) instrument [24], distributed to 14 respondents consisting of active Gen Z users of the X platform. The system obtained an average SUS score of 80 (SD = 12.93, 95% CI: 73-87), placing

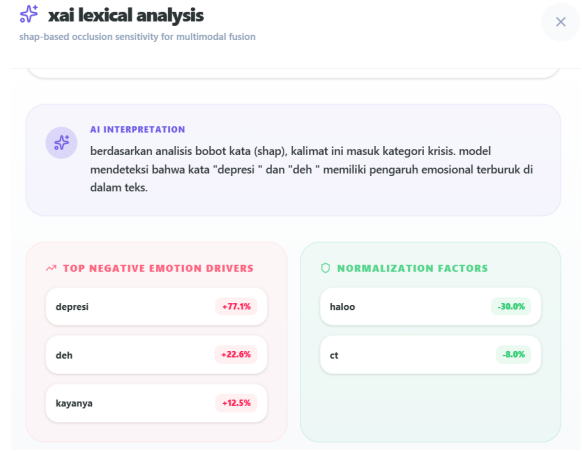


Fig. 4. SHAP attention map. The red highlighted texts represent Indonesian words that strongly influence the “Indicated” prediction.

it in the *Acceptable* category with an Excellent adjective rating. According to the SUS adjective rating scale, a score of 80 or above is classified as Excellent, indicating a high level of user satisfaction and learnability. This proves that the browser extension and analytical dashboard successfully isolated the heavy computational AI complexity on the backend while presenting a lightweight, intuitive visual interface operable without technical assistance.

E. Ethical and Privacy Considerations

Given the sensitive nature of inferring psychological distress from personal digital footprints, several ethical safeguards and privacy preservation protocols were integrated into the system design:

- *Data Anonymization and Zero Retention*: All user identifiers (including @usernames, account IDs, and media URLs) are stripped during client-side DOM extraction. The FastAPI backend processes inferencing strictly in volatile memory without retaining raw user posts or personal data.
- *Ethical Output Framing*: To prevent nocebo effects, anxiety, and inaccurate self-diagnosis, the system deliberately displays risk levels as non-diagnostic “Stress Indications” accompanied by prominent disclaimers rather than psychiatric verdicts.
- *Non-Intervention and Supportive Guidance*: The system adheres to a strict non-intervention principle, operating passively without unsolicited automated messaging to analyzed accounts. Instead, structured DSM-5 symptom mapping and basic supportive guidance are provided exclusively on the dashboard as preliminary reference for clinical decision support.

V. CONCLUSION AND SUGGESTION

This study successfully developed an early detection expert system for potential depression based on Multimodal Transformers (text and images), implemented as a browser

extension. Although the Late Fusion between IndoBERTweet and ViT achieved 82.10% accuracy, the Paired t-Test results proved that the image modality did not provide a statistically significant performance improvement. Nevertheless, the multimodal fusion was retained because it functionally suppressed False Negatives to the lowest count among all multimodal variants (13 cases vs. 16 for ResNet-18), a metric considered extremely vital in psychological diagnostic systems.

Contextualized against related work, this study's approach offers distinct contributions. Compared to Faiz et al. [11], whose CNN-BiGRU pipeline targets post-level multimodal detection, this study operates at the user-level by aggregating temporal tweet histories, more closely mirroring the DSM-5 diagnostic window. Compared to Al Asad et al. [9] and Zogan et al. [10], who achieve user-level modeling through behavioral features alone, this study integrates visual semantics and provides clinical-grade transparency through XAI, a dimension absent in prior work.

This system transcended the traditional detection paradigm (post-level analysis) toward comprehensive temporal understanding at the user level (user-level analysis). With the Hybrid Scoring method and a 14-day temporal monitoring window in accordance with DSM-5 guidelines, the system effectively distinguishes fluctuating situational emotions from consistent clinical depressive patterns. Dual validation (clinically through Expert Judgment and by end-users via UAT, SUS=80, Acceptable/Excellent) confirms the system is viable as an early warning system bridge. Transparency through Explainable AI (XAI) ensures every detection has intuitive visual accountability.

Several directions remain open for future development. At the model level, improvements include: more precise language filtering using FastText LangDetect to reduce code-mixing noise; an expanded multimodal dataset with stricter sarcasm annotation protocols; and exploration of additional visual modalities such as audio and video from TikTok and X Spaces to capture a richer spectrum of emotional expression.

At the system level, future work will focus on incorporating temporal behavioral pattern signals such as active-hour distribution as an additional depression indicator, and transitioning from browser-based DOM extraction to the official X API for a more ethical and non-intrusive data collection pipeline. Real-time deployment on mobile platforms is also planned to broaden the system's clinical reach and accessibility across diverse demographic groups.

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